

4.5.1 Forces and their interactions

4.5.1.1 Scalar and vector quantities

Content	Key opportunities for skills development
Scalar quantities have magnitude only. Vector quantities have magnitude and an associated direction. A vector quantity may be represented by an arrow. The length of the arrow represents the magnitude, and the direction of the arrow the direction of the vector quantity.	

4.5.1.2 Contact and non-contact forces

Content	Key opportunities for skills development
A force is a push or pull that acts on an object due to the interaction with another object. All forces between objects are either: • contact forces – the objects are physically touching • non-contact forces – the objects are physically separated. Examples of contact forces include friction, air resistance, tension and normal contact force. Examples of non-contact forces are gravitational force, electrostatic force and magnetic force. Force is a vector quantity. Students should be able to describe the interaction between pairs of objects which produce a force on each object. The forces to be represented as vectors.	

4.5.1.3 Gravity

Content	Key opportunities for skills development
Weight is the force acting on an object due to gravity. The force of gravity close to the Earth is due to the gravitational field around the Earth. The weight of an object depends on the gravitational field strength at the point where the object is.	

Content	Key opportunities for skills development
The weight of an object can be calculated using the equation: weight = mass × gravitational field strength [$W = m g$] weight, W, in newtons, N mass, m, in kilograms, kg gravitational field strength, g, in newtons per kilogram, N/kg (In any calculation the value of the gravitational field strength (g) will be given.) The weight of an object may be considered to act at a single point referred to as the object's 'centre of mass'.	MS 3b, c Students should be able to recall and apply this equation.
The weight of an object and the mass of an object are directly proportional. Weight is measured using a calibrated spring-balance (a newtonmeter).	MS 3a Students should recognise and be able to use the symbol for proportionality, $\propto$

4.5.1.4 Resultant forces

Content	Key opportunities for skills development
A number of forces acting on an object may be replaced by a single force that has the same effect as all the original forces acting together. This single force is called the resultant force. Students should be able to calculate the resultant of two forces that act in a straight line. (HT only) Students should be able to: describe examples of the forces acting on an isolated object or system	
use free body diagrams to describe qualitatively examples where several forces lead to a resultant force on an object, including balanced forces when the resultant force is zero. (HT only) A single force can be resolved into two components acting at right angles to each other. The two component forces together have the same effect as the single force.	WS 1.2
(HT only) Students should be able to use vector diagrams to illustrate resolution of forces, equilibrium situations and determine the resultant of two forces, to include both magnitude and direction (scale drawings only).	MS 4a, 5a, b